Pablo Rocabert Messa

Urban Data Scientist & Urban Designer

Geospatial Analytics · City Digital Twins · Urban Planning · Architecture

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EU citizen (Spain) · Eligible to work across the EU/EEA · Open to relocation and hybrid roles

PROFILE

Architect and urban planner turned urban data scientist, with 8 years of experience across architecture, real estate investment analysis and urban analytics, including 2+ years building city digital twins at Aretian. I turn cadastral, zoning, socio-economic and climate data into parcel-level evidence and design proposals for city councils, public administrations and real estate clients. Hands-on with Python and GIS data pipelines, statistical modelling, LLM-assisted data classification and computational design, backed by a Master's in Data Analytics & AI.

CORE SKILLS

Geospatial & data	Python (pandas, GeoPandas, Shapely, NumPy, scikit-learn, Rasterio) · QGIS · ArcGIS Pro · PostGIS · GDAL/OGR · GeoPackage · INSPIRE / GML · OGC WFS/WMS · OpenStreetMap / Overpass API · Folium
Analytics & AI	ETL data pipelines · Data cleaning & QA · Regression and hedonic price modelling · KPI frameworks · LLM-assisted classification (Anthropic API) · Data visualisation (maps, Sankey diagrams)
Urban planning	Urban & territorial planning · Land use & zoning analysis · Cadastral analysis · Housing · Mobility & accessibility · Climate resilience (flood, heat, SUDS) · Digital twins
Design & delivery	Rhino · Grasshopper · AutoCAD · Revit · SketchUp · 3ds Max · V-Ray · Unity · Excel investment models · Project & team coordination · Stakeholder reports and presentations

PROFESSIONAL EXPERIENCE

Aretian Urban Analytics and Design — Barcelona, Spain

Urban analytics and city-science consultancy (Barcelona · Cambridge, MA) building city digital twins for public- and private-sector clients.

Senior Urban Designer & Data Scientist — Aug 2025 – Present

Urban Designer & Data Scientist — Aug 2024 – Jul 2025

- Build Python and QGIS data pipelines (extraction, cleaning, spatial joins, QA) to analyse **housing, land use, mobility and economic activity** at parcel and metropolitan scale.
- Contribute to the **City Digital Twin of the Barcelona Metropolitan Region**, linking each parcel to what can be built and how much, to identify development and transformation opportunities.
- Lead analytical workstreams end to end, from data sourcing and modelling to client-ready **maps, reports and presentations** for city councils, public bodies and private clients in Spain and South Africa.
- Develop **urban design proposals in Rhino/Grasshopper** grounded in territorial analysis, and coordinate architecture and data teams on scope, timelines and deliverables.
- Promoted to Senior in August 2025, one year after joining.

Turull Sorensen Arquitectura — Barcelona, Spain

Architect — Mar 2022 – Jul 2024

- Delivered basic and construction-stage design projects for residential and commercial refurbishments and new builds: design definition, construction details and drawings.
- Reviewed zoning and building regulations and wrote technical reports defining project conditions and compliance.
- Coordinated technical documentation with public authorities, developers, consultants and contractors through permitting and construction.

Busquets Gálvez — Barcelona, Spain

Real Estate Analyst — Oct 2018 – Mar 2022

- Built Excel investment models (costs, revenues, returns) and analysed planning and cadastral conditions to support acquisition and development decisions.
- Valued a hotel and a residential building in Porto (Portugal) and supported office-market analysis in Barcelona's 22@ innovation district.

Jorge Vidal Studio — Barcelona, Spain

Architecture Intern — May 2018 – Oct 2018

- Supported design and documentation with 3D models, visualisations and parametric design studies.

SELECTED PROJECTS

Catalonia parcel-level real estate dataset — Aretian · 2026

Built a five-phase pipeline producing one GeoPackage for **all of Catalonia** that enriches Catastro INSPIRE parcels and buildings with MUC zoning classification, building use and business-activity flags (spatial joins with distance thresholds), so developers can screen **current vs. developable activity** parcel by parcel.

Stack: Python · GeoPandas · pyogrio · Catastro INSPIRE (CP/BU, GML) · MUC zoning · GeoPackage (EPSG:25831)

Hedonic housing-price model — Aretian · Catalan City Digital Twin

Modelled residential prices with **OLS regression on log-transformed variables** to explain price drivers and feed the housing workstream of the Catalan digital twin.

Stack: Python · pandas · regression modelling

Flood risk, green infrastructure and climate resilience — Aretian · Bessó Digital Metropolità de Barcelona

Designed a framework for the digital twin to **simulate flood risk and flag candidate parcels for SUDS** and water-retention systems, using ACA and ICGC layers; authored **10 climate-resilience recommendations (7 heat, 3 flood)** for a Puig Foundation project.

Stack: QGIS · GeoPandas · OWSLib · ogr2ogr · OGC WFS/WMS · Copernicus / AMB / Open Data BCN

Western Cape metropolitan innovation diagnostic — Aretian · South Africa

Structured urban, economic and research data for **11 municipalities** (incl. Cape Town and Stellenbosch): NAICS-classified business data, university research output (UCT, Stellenbosch, UWC, CPUT) and KPIs such as Revealed Comparative Advantage; wrote the midway and progress reports.

Stack: Python · QGIS · GeoServer WMS · Dun & Bradstreet data · KPI frameworks

Innovation-ecosystem pipeline: Harvard & MIT — Aretian · USA

Built an **LLM-assisted company-classification pipeline** and a three-stage Sankey model (researchers → innovative industries → sectors) with a custom crosswalk across changing flow units; extracted campus building footprints from ArcGIS FeatureServer and OpenStreetMap.

Stack: Python · Anthropic API · Sankey diagrams · Overpass API · ArcGIS REST

EU European Urban Initiative bid: municipal digital twin — Aretian · Sant Joan Despí, 2026

Contributed the digital-twin component of a consortium bid for EU urban-innovation funding alongside SATEC and the Barcelona Supercomputing Center (BSC).

Stack: Proposal writing · Digital twins · Public-sector innovation

EDUCATION

Master's in Data Analytics & AI — Nov 2025 – May 2026
Nuclio Digital School, Barcelona · Python, data analysis, artificial intelligence

Master's in Architecture and Urbanism — 2023 – 2024
La Salle – Universitat Ramon Llull, Barcelona

Bachelor's Degree in Architecture — 2014 – 2023
Universitat Politècnica de Catalunya (UPC), Barcelona

LANGUAGES

Spanish (native) · Catalan (native) · English (professional working proficiency) · French (basic)

Portfolio and project references available on request.